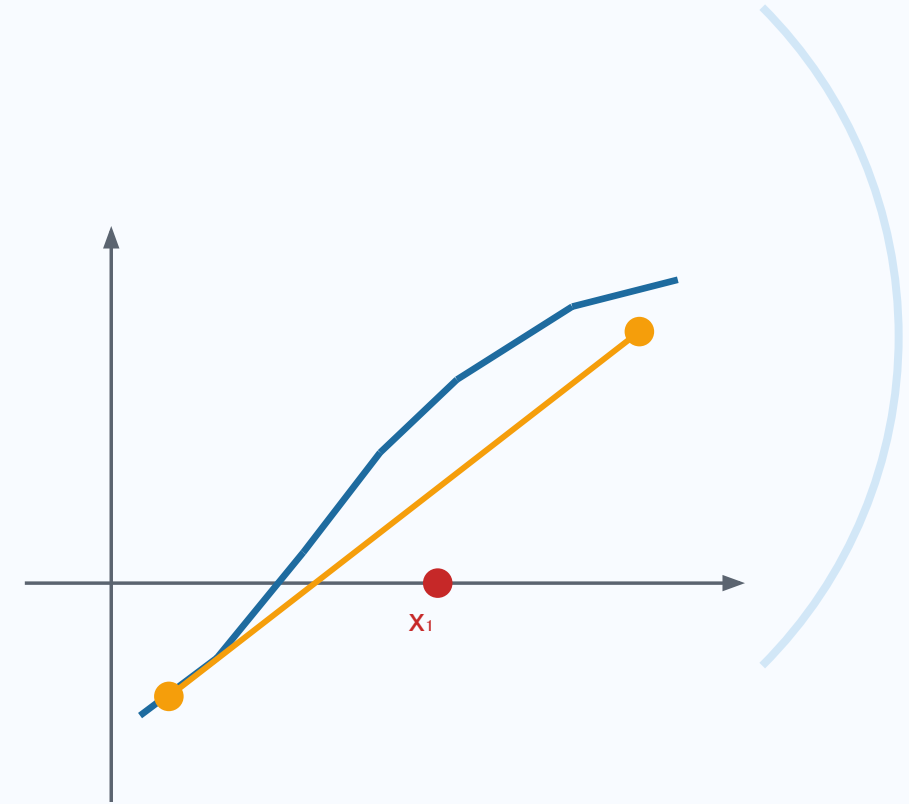


Method of False Position

(Regula-Falsi Method)

Finding the real root of $f(x) = 0$ using a straight chord between two points.



Chord intercept $\approx$ root

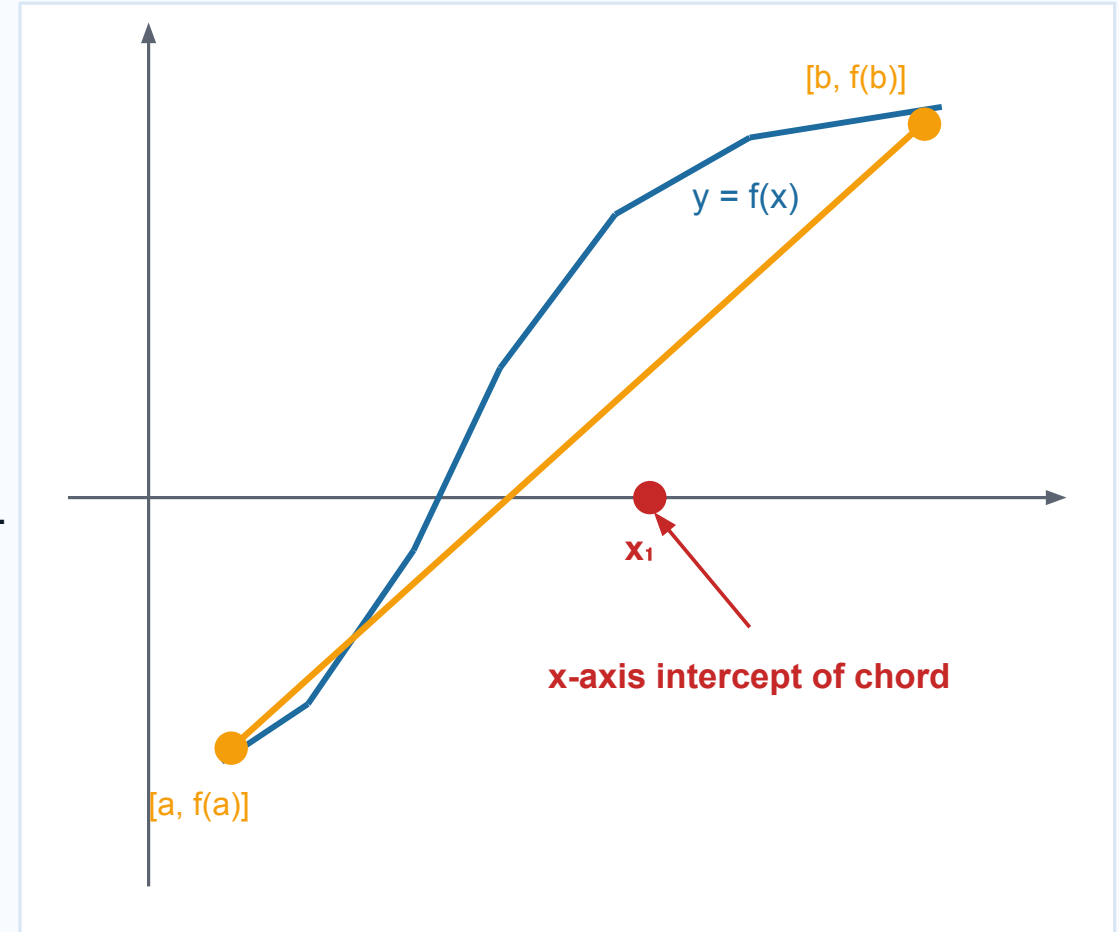
Core Idea

Use a chord instead of the actual curved function.

Need opposite signs

- Choose a and b so that $f(a)$ and $f(b)$ have opposite signs.
- This guarantees the root lies between a and b .
- Draw a straight line from $[a, f(a)]$ to $[b, f(b)]$.
- Where this line cuts the x -axis gives the next estimate.

The method repeats until the estimate is accurate enough.



Formula and Update Rule

Calculate the chord intercept, then keep the interval that still contains the root.

First approximation:

$$x_1 = a - [f(a)(b - a) / (f(b) - f(a))]$$

Same formula can also be written as:

$$x_1 = [a f(b) - b f(a)] / [f(b) - f(a)]$$

Then check the sign:

If $f(a)$ and $f(x_1)$ have opposite signs:

root is in $[a, x_1]$ → replace b by x_1

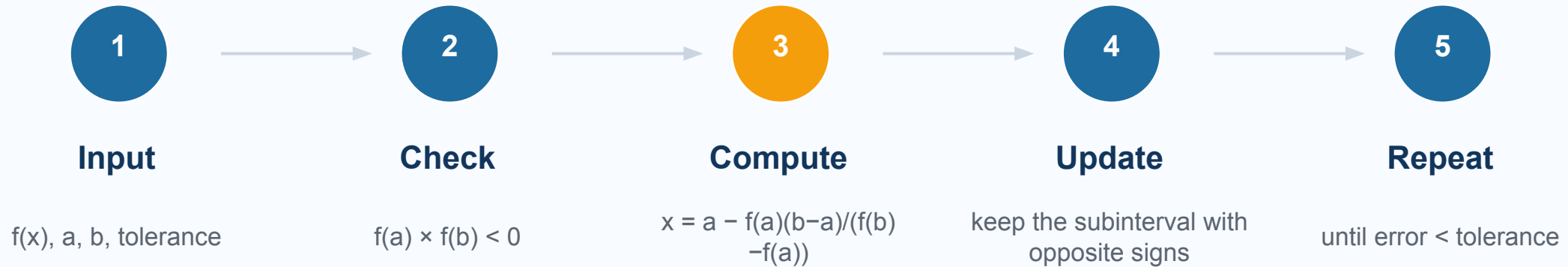
Otherwise:

root is in $[x_1, b]$ → replace a by x_1

Stop when the error is small enough, e.g., $|x_{\text{new}} - x_{\text{old}}| < \text{tolerance}$.

Algorithm: False Position Method

A simple step-by-step procedure.



Pseudocode while error $>$ tolerance: compute x ; if $f(a)f(x) < 0$ then $b = x$; else $a = x$; repeat

Output: the final x value is the approximate root.

Quick Example

Find the root of $f(x) = x^2 - 4$ in the interval $[1, 3]$.

Initial values:

$$a = 1, b = 3$$

$$f(1) = -3 \text{ and } f(3) = 5$$

Opposite signs $\Rightarrow$ root exists between 1 and 3.

First estimate: $x_1 = 1 - [-3(3-1)/(5-(-3))] = 1.75$

Iter.	a	b	x	f(x)
1	1.000	3.000	1.750	-0.938
2	1.750	3.000	1.941	-0.235
3	1.941	3.000	1.986	-0.056
4	1.986	3.000	1.997	-0.014

The estimates move toward the actual root: $x = 2$.

False Position vs. Bisection: Bisection uses the midpoint; False Position uses the chord intercept.